Exercise sheet: Week 1 - Solutions

1. (a) $\overline{X} \sim \text{N}(\mu, {}^{\sigma^2}/_n)$, so $\mathbf{E}(\overline{X}) = \mu$ and $B(\overline{X}) = 0$. Therefore $\overline{X}$ is an unbiased estimator for μ. $MSE(\overline{X}) = \mathbf{Var}(\overline{X}) + B(\overline{X})^2 = \mathbf{Var}(\overline{X}) = \frac{\sigma^2}{n} \to 0$ as $n \to \infty \Rightarrow \overline{X}$ is consistent for μ.

   (b) For a random sample from *any* distribution with finite mean μ and finite variance σ^2, $\overline{X}$ has mean μ and variance σ^2/n. The result of part (a) generalises to any such distribution.

2. Beta$(\theta, 1)$ has mean $\frac{\theta}{\theta+1}$. To get the MME, we solve $\bar{X} = \frac{\theta}{\theta+1}$ to get $\widetilde{\theta} = \frac{\bar{X}}{1-\bar{X}}$. (NB: in this example it is not easy to get $\mathbf{E}(\widetilde{\theta})$ and $\mathbf{Var}(\widetilde{\theta})$!)

3. For $X \sim \text{N}(\mu, \sigma^2)$, $\mathbf{E}(X) = \mu$ and $\mathbf{E}(X^2) = \sigma^2 + \mu^2$. To get MMEs of μ and σ^2, we solve

$$\begin{aligned} \bar{X} &= \mu \\ \tfrac{1}{n}\textstyle\sum_i X_i^2 &= \sigma^2 + \mu^2 \end{aligned}$$

   to get

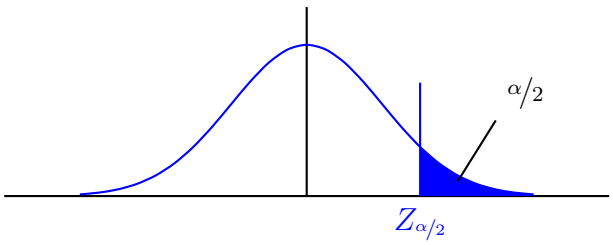
$$\widetilde{\mu} = \bar{X}$$

$$\begin{aligned} \widetilde{\sigma}^2 &= \tfrac{1}{n}\textstyle\sum_i X_i^2 - \bar{X}^2 \\ &= \tfrac{1}{n}\left(\textstyle\sum_i X_i^2 - n\bar{X}^2\right) \\ &= \tfrac{1}{n}\textstyle\sum_i \left(X_i^2 - \bar{X}^2\right) \\ &= \tfrac{1}{n}\textstyle\sum_i \left(X_i^2 + \bar{X}^2 - \bar{X}^2 - \bar{X}^2\right) \\ &= \tfrac{1}{n}\textstyle\sum_i \left(X_i^2 - 2\bar{X}^2 + \bar{X}^2\right) \\ &= \tfrac{1}{n}\textstyle\sum_i \left(X_i^2 - 2X_i\bar{X} + \bar{X}^2\right) \\ &= \tfrac{1}{n}\textstyle\sum_i (X_i - \bar{X})^2 \end{aligned}$$

   as required.

4. Recall that $\sum_i X_i = n\bar{X}$ and $\bar{X} \sim \text{N}(0, \frac{\theta}{n})$. Hence $\frac{\bar{X}-0}{\sqrt{\theta/n}} \sim \text{N}(0,1)$ and $Z^\star = \frac{\sum_i X_i}{\sqrt{n\theta}} \sim \text{N}(0,1)$.

   Using $Z^\star$ as a pivot: $\mathbf{P}(-Z_{\frac{\alpha}{2}} < \frac{\sum_i X_i}{\sqrt{n\theta}} < Z_{\frac{\alpha}{2}}) = 1 - \alpha$.



   Rearrange these inequalities to get

$$\frac{\left(\sum_i X_i\right)^2}{n\theta} < Z^2_{\alpha/2} \qquad \Rightarrow \qquad \theta > \frac{\left(\sum_i X_i\right)^2}{nZ^2_{\alpha/2}}.$$

   Comment: This is not a particularly sensible CI: it is always 1-sided.

Now consider $Z = \frac{(\sum_i X_i)^2}{n\theta} \sim \chi^2_1$, which can be used as a pivot:

$$\mathbf{P}\left(\chi^2_1\left(100\left(1-\frac{\alpha}{2}\right)\right) < \frac{(\sum_i X_i)^2}{n\theta} < \chi^2_1\left(100\frac{\alpha}{2}\right)\right) = 1-\alpha$$

Rearrange to get

$$\frac{(\sum_i X_i)^2}{n\chi^2_1\left(100\frac{\alpha}{2}\right)} < \theta < \frac{(\sum_i X_i)^2}{n\chi^2_1\left(100\left(1-\frac{\alpha}{2}\right)\right)}$$

as a $100(1-\alpha)\%$ CI. This should give a *more appropriate* CI.

5. $X_1, X_2, \ldots, X_n \sim \mathrm{U}(0,\theta)$ and $X_{(n)} = \max\{X_i\}_{i=1}^n$.

   From the hint, $Z = \frac{X_{(n)}}{\theta} \sim \mathrm{Beta}(n,1)$. Now, the pdf of Z is $f(z) = nz^{n-1}$, $0 < Z < 1$, and cdf of Z is $F(z) = z^n$.

   We seek a and b such that $F(a) = 0.025$, $F(b) = 0.975 \;\Rightarrow\; a = (0.025)^{\frac{1}{n}}, b = (0.975)^{\frac{1}{n}}$. Therefore

$$\mathbf{P}\left((0.025)^{\frac{1}{n}} < \frac{X_{(n)}}{\theta} < (0.975)^{\frac{1}{n}}\right) = 0.95\,.$$

   Rearrange this to obtain $\frac{X_{(n)}}{(0.975)^{\frac{1}{n}}} < \theta < \frac{X_{(n)}}{(0.025)^{\frac{1}{n}}}$ as the 95% CI.

   Data: $n = 5$, $X_{(5)} = 1.3$, $a = 0.478$, $b = 0.995$. These give the CI $\left(\frac{1.3}{0.995}, \frac{1.3}{0.478}\right)$ or $(1.31, 2.72)$.

6. $\underline{X}$ from $\mathrm{Poi}(\theta) \;\Rightarrow \sum_i X_i \sim \mathrm{Poi}(n\theta) \approx \mathrm{N}(n\theta, n\theta)$.

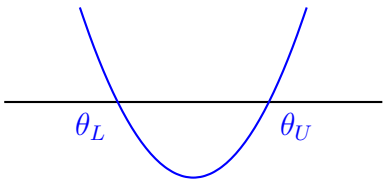
   Use $\frac{\sum_i X_i - n\theta}{\sqrt{n\theta}} = \frac{\bar{X}-\theta}{\sqrt{\theta/n}} \approx \mathrm{N}(0,1)$ as a pivot:

$$\mathbf{P}\left(-Z_{\alpha/2} < \frac{\bar{X}-\theta}{\sqrt{\theta/n}} < Z_{\alpha/2}\right) = 1-\alpha\,.$$

   Square to rearrange the inequality: $\left(\frac{\bar{X}-\theta}{\sqrt{\theta/n}}\right)^2 < Z^2_{\alpha/2}$. This gives

$$\theta^2 - \left(2\bar{X} + \frac{Z^2_{\alpha/2}}{n}\right)\theta + \bar{X}^2 < 0\,,$$

   so the CI (θ_L, θ_U) comes from the root of this quadratic:



   (use the quadratic formula)

A much easier approach is to replace θ by $\hat{\theta} = \overline{X}$ in the denominator:

$$\frac{\overline{X} - \theta}{\sqrt{\overline{X}/n}} \approx \mathrm{N}(0,1) \quad \text{leading to} \quad \overline{X} \pm Z_{\alpha/n}\sqrt{\frac{\overline{X}}{n}}$$

as $100(1-\alpha)\%$ CI for θ.